

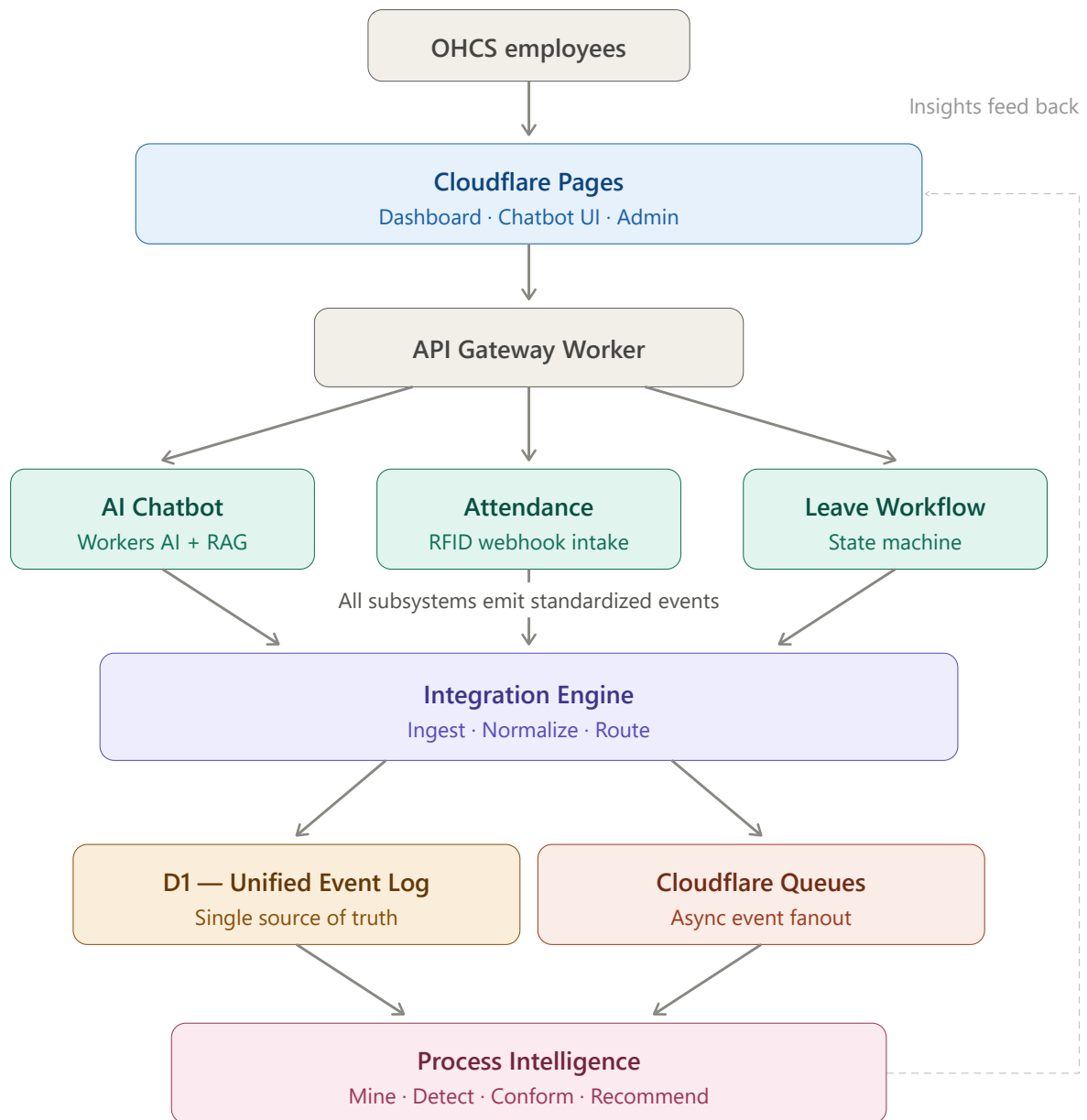
IIE Demonstration Guide

For: Student demonstrating the Intelligent Integration Engine (IIE) to a project supervisor **System:** OHCS Process Intelligence Platform — Cloudflare Workers + D1 + Workers AI + Vectorize, React dashboard **Time needed:** 20–30 minutes for the full walkthrough

1. The 30-second pitch (memorise this)

"IIE is a process intelligence platform for the Office of the Head of Civil Service. It collects a unified event log from three source systems — RFID attendance, the leave workflow, and an HR chatbot — and automatically mines that log to discover how processes actually run: it draws the real process map, finds bottlenecks, flags cases that skip required approval steps, and turns all of that into decision-support recommendations for management. Staff get self-service leave with the real OHCS approval chains built in — standard leave through F&A, study leave through RTDD — and an AI assistant that answers HR policy questions and can submit leave requests in plain language."

Key phrase if asked what makes it different from a normal dashboard: **"A normal dashboard shows you numbers you asked for. Process mining discovers the actual workflow from the event log and tells you where reality deviates from policy — without anyone configuring it."**



System architecture — source subsystems publish canonical events into the unified log; mining and the dashboard consume it.

2. Pre-demo checklist (do this 30 minutes before)

You need: the project folder open, internet access (the AI features call live Cloudflare services even in local dev), and one terminal.

```
cd "C:/dev/IIIE - Nii"  
npm run dev           # starts the Worker + dashboard on http://localhost:8787
```

Then verify, in a second terminal:

```
curl http://localhost:8787/health      # expect: {"status":"ok"} (or similar 200 response)  
curl http://localhost:8787/api/stats/overview
```

The overview should show roughly: **150 employees, ~33,300 events, ~4 flagged bottlenecks**. If it shows 0 events, the database is empty — run `npm run seed` (takes ~3 minutes) before the demo.

Open **http://localhost:8787** in your browser and click through all four tabs once to warm everything up. Keep this browser window ready. (Tabs are hash-routed — refreshing or sharing `http://localhost:8787#intelligence` lands directly on that tab.)

If you ever need fresh data: `npm run seed -- --reset` wipes every table and re-seeds in one step (plain `npm run seed` is additive and duplicates events).

3. Know your data (the story behind the numbers)

The demo dataset simulates **6 months** of OHCS operations. Understanding it lets you answer "why does it show that?" confidently.

The organisation (real OHCS structure)

- **9 organisational units**, keyed by their real acronyms:
- 5 directorates: **RSIMD** (Research, Statistics & Information Management), **PBMED** (Planning, Budgeting, Monitoring & Evaluation), **CMD** (Career Management — where HR sits), **F&A** (Finance & Administration), **RTDD** (Recruitment, Training & Development)
- 4 units: **RCU** (Reform Coordinating Unit), **CSC** (Civil Service Council), **IAU** (Internal Audit Unit), **PR** (Public Relations Unit)
- **150 staff in total** (the full OHCS staff strength), following the civil-service grading structure: every unit has a **Director** (management), a **Deputy Director** and an **Assistant Director I** (middle management), plus **122 officers** of lower grades distributed across the units. One **HR officer** sits in CMD.
- Two deputy directors carry leave-administration designations: F&A's Deputy Director is the **admin officer** verifying standard leave; RTDD's Deputy Director is the **Schedule Officer** reviewing study leave.

The three event sources (~33,300 events total)

SOURCE	EVENTS	WHAT IT IS
ATTENDANCE	~31,700	RFID clock-in / clock-out taps, weekdays over 6 months
LEAVE_WORKFLOW	~1,200	Full leave-approval chains (256 cases)
CHATBOT	~400	Staff queries to the HR assistant

The prescribed leave workflows (policy)

Standard leave (annual, sick, maternity, casual) — administered at OHCS by F&A: `leave_submitted` → `supervisor_review` → `fa_verification` → `director_fa_approval` → `completed`

Study leave (OHCS and the entire Ghana Civil Service) — administered centrally by RTDD: `leave_submitted` → `supervisor_review` → `rtdd_review` → `director_rtdd_approval` → `completed`

- **Supervisor review** — the officer's middle-management supervisor in their own unit (Assistant Director / Deputy Director tier)
- **F&A verification / RTDD review** — the administering directorate's officer (F&A admin officer, or the RTDD Schedule Officer)
- **Director approval** — the Director of that directorate (**Director F&A**, or **Director RTDD** for study leave)

What the seed deliberately "planted" (this is what the system should find)

1. **A slow step:** F&A verification is intentionally slow — the transition `supervisor_review` → `fa_verification` has a **median of 3.3 days (P95: 5.7 days)**. The bottleneck detector should flag exactly this.
2. **Policy violations:** ~10% of leave cases **skip supervisor review entirely** (they jump straight to verification). The conformance checker should catch **all 22 of them**.

3. **Department outliers:** CSC has the worst punctuality (**13.8% late**); the Public Relations Unit has the slowest leave-approval cycle (**7.6 days**).

This planted ground truth is important: it means you can *prove* the mining works, because you know the right answers in advance.

4. Demo script — Act 1: Operations tab (3–4 min)

Narrative: *"This is the live operations view — what's happening across the office right now."*

Walk through top to bottom:

1. **System map** (top): shows the architecture — source subsystems (Attendance, Leave Workflow, Chatbot) feeding the Integration Engine, which feeds Process Intelligence. *"Every subsystem publishes events in one canonical format into a unified event log — that's the core design decision. New systems just start emitting events; nothing else changes."*
 2. **Stat cards:** 150 employees, open leave requests. *"These poll the API every 10 seconds — if an event arrives, the numbers move on their own."*
 3. **Live event feed** (left): the 25 most recent events, colour-coded by source (green = attendance, indigo = leave, amber = chatbot). *"This feed is pushed live over server-sent events — the moment any subsystem posts, it appears here. And every row is clickable."* **Click a row** — the case's full chronological event trace opens in a modal. *"Every case carries its complete audit trail."*
 4. **Attendance heatmap** (right): 30 days, green intensity = clock-in volume, red tint = days where >25% of staff were late. **Hover a red-tinted cell** to show the tooltip (e.g. "48 in, 9 late, 2 missing out"). *"You can spot bad Mondays at a glance."*
 5. **Leave pipeline:** counts of cases waiting at each stage — supervisor review, F&A verification, director F&A approval, RTDD review, director RTDD approval, completed, rejected, escalated. *"This is the state machine working — every leave request moves through one of the two prescribed chains, and the pipeline shows where work is piling up."*
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5. Demo script — Act 2: Process Intelligence tab (5–6 min) — the centrepiece

Narrative: *"This is where the platform earns its name. Nobody drew this process map — the system discovered it from the event log."*

1. **The discovered process map** (top): - Point at the nodes and edges: *"Each node is an activity, each arrow is an observed hand-off with a case count and median transition time."* - **The two chains moment:** *"Look — the miner discovered BOTH prescribed chains on its own: standard leave flowing through F&A verification and Director F&A approval, and study leave flowing through RTDD review and Director RTDD approval. Nobody told it about the F&A/RTDD split — it read it out of the data."* - Point at the **red edges/pills**: *"Red means the transition is over its SLA threshold. The map overlays bottleneck analysis directly onto the discovered process."* - Mention the **rejected** branch (red-accented node): *"It also discovered the rejection paths on its own."* - Source switcher: click **ATTENDANCE** and **CHATBOT** briefly to show it mines all three sources, then return to LEAVE_WORKFLOW.
2. **Bottlenecks panel** (bottom left): - Read the top row: `supervisor_review -> fa_verification` — **median 3.3d, P95 5.7d**, flagged red. - *"The single slowest hand-off in the leave process is getting from the supervisor's desk to F&A verification — a median of over three days. That's a staffing or routing problem at one specific step, and the system localised it exactly."*
3. **Workflow variants** (bottom right): - *"256 leave cases collapse into a handful of distinct paths — the full F&A chain, the full RTDD study chain, the rejection variants... and these — `leave_submitted → fa_verification → ...` — where supervisor review never happened."*
4. **Conformance panel:** - *"The conformance checker compares every case against the prescribed model for its leave type. It found 22 cases that bypassed supervisor review — all type `skipped_step`. In a real deployment this is the audit list."* - Show the conformant-rate bar (~91%). - **Click a case id** — the full trace opens, showing the jump from `leave_submitted` straight to verification. *"Every finding is inspectable down to the raw events."*

If the supervisor asks "how do you know it's right?" — that's your cue: *"The demo data deliberately plants exactly these problems — a known slow step and 22 known violations — and the validation harness measures detection against that ground truth."* (Then show Act 6 if time permits.)

6. Demo script — Act 3: Decision Support tab (3–4 min)

Narrative: *"Mining results are for analysts. This page is for management — it turns the analysis into plain-English recommendations, ranked by severity."*

1. **Top insight banner** (dark card): the highest-severity finding — the slow `supervisor_review` → `fa_verification` step, with a concrete suggestion (backup approver / SLA alerts).
 2. **Department comparison** (the organisational-payoff moment): - *"Every unit, compared on two fixed scales: late-arrival rate and leave-approval cycle. Red marks anything more than 1.5× the office average."* - Point at **Civil Service Council — 13.8% late** (worst punctuality) and **Public Relations Unit — 7.6-day leave cycle** (slowest). - *"This is the kind of cross-directorate comparison OHCS currently can't produce without manual spreadsheets — here it's a by-product of the event log."*
 3. **Recommendation cards**: bottleneck alerts, the conformance finding ("22 cases bypassed supervisor review — enforce routing at submission"), and the variability note.
 4. **The export buttons** (top right): *"And this page leaves the building — **Download CSV** drops the recommendations and the department table straight into Excel, and **Print / Save as PDF** produces a clean one-pager for a director's memo."*
 5. Mention the roadmap line honestly: *"The recommendations are rule-generated today; an AI narrative layer over them is a planned next phase."*
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7. Demo script — Act 4: the AI assistant (4–5 min)

Click **"Ask OHCS assistant"** (the gradient pill, bottom-right, on every page). The panel shows who's speaking via the **"Speaking as"** dropdown — *"the assistant knows who it's talking to; per-user sign-in arrives with the SSO phase."* The empty state offers **clickable suggestion chips** — tapping one asks it instantly (a handy shortcut mid-demo).

Narrative: *"The assistant is retrieval-augmented: it embeds the actual OHCS policy documents into a vector database and answers strictly from retrieved excerpts — it can't invent policy."*

Ask these in order (expected answers in brackets — the policy corpus guarantees them):

1. **"How many days of annual leave am I entitled to?"** → 30 working days. *Point at the indigo source chips under the reply — proof it's RAG over the real policy documents, not memorised.*
2. **"What is the grace period for morning clock-in?"** → 30 minutes (late after 8:30 a.m.).
3. **"Who approves study leave applications?"** → the Director, RTDD — after review by the RTDD Schedule Officer. *"That's the Study Leave Policy we just saw in the process map — the chatbot and the miner agree because both read the same reality."*
4. **"Do I need a medical certificate for two days of sick leave?"** → No — certificates only beyond 3 consecutive days.
5. **The personal-data moment: "How many days was I late this month?"** → answered from the *employee's own attendance records*, no LLM involved. *"This is the hybrid design: personal data queries go straight to the database through deterministic rules — the language model only handles policy language."*
6. **The transaction moment: "I'd like to request annual leave from 3 August 2026 to 7 August 2026."** - The assistant confirms: *"Done — your annual leave request (2026-08-03 to 2026-08-07) is submitted and now with your supervisor for review."* - **Then prove it's real:** flip to the **Operations tab** — the new `leave_submitted` event is already at the top of the live feed (pushed over SSE), and the pipeline's `supervisor review` count went up by one. - *"The chatbot didn't just talk — it executed a real workflow transaction against the state machine, and you watched the event land in the log."*

Safety point worth volunteering: *"Leave dates are only ever taken from the user's own message, never model-generated — the LLM can't invent dates for a transaction."*

8. Demo script — Act 5: My Leave (4–5 min) — the workflow made tangible

Narrative: *"The chatbot is one front door to the leave process. This is the other — a self-service portal where officers submit and track leave, and approvers act on it. Same state machine underneath."*

1. **Submit:** with "Acting as" set to any officer, pick *annual*, choose dates, **Submit**. The request appears under **My requests**, *pending · waiting at supervisor review*.
 2. **Approve the full chain in the Approver inbox** (right side): - **Supervisor review** — pick the officer's unit Assistant Director I from the dropdown (*"the picker only offers officers whose role and unit allow them to act — the state machine would reject anyone else"*) → **Approve**. The request moves to **F&A verification**. - **F&A verification** — the picker now only offers the F&A admin officer (the F&A Deputy Director) → **Approve**. Moves to **director F&A approval**. - **Director F&A approval** — only Director F&A is offered → **Approve**. The request lands in **completed**. - Flip back to **Operations**: the whole chain just streamed through the live feed.
 3. **Show the routing rule:** submit a **study** request, approve at supervisor review — and point at the inbox: *"Study leave doesn't go to F&A. It routes to RTDD — the Schedule Officer reviews, and only Director RTDD can approve. That's the real Civil Service rule, enforced by the same engine the miner audits against."*
 4. **Track:** expand a request under My requests — the step-by-step timeline with actor and timestamp. *"Every officer can see exactly where their request is and whose desk it's on."*
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9. Demo script — Act 6 (optional, for technical supervisors): proof it works (3 min)

In a terminal, with the dev server running:

```
npm run validate
```

This measures the project's six PRD success metrics live, against the planted ground truth. All six pass:

METRIC	TARGET	TYPICAL RESULT
Event capture latency	< 500 ms	p50 ~8 ms
Workflow variants discovered	≥ 3	9
Bottleneck detection	F&A-verification transition flagged	<code>supervisor_review</code> → <code>fa_verification</code> , median 3.3d
Conformance detection rate	> 80% of known violations	22/22 = 100%, 0 false positives
Dashboard load time	< 2 s	< 100 ms
Chatbot policy resolution	> 60%	5/5 = 100%

If they want more: `npm test` runs the 32-test suite (ingestion, the leave state machine — including the study-leave RTDD routing and F&A scoping, DFG miner, conformance checker, SSE stream, full mining pipeline) inside the real Workers runtime.

You can also show raw API output, e.g. `http://localhost:8787/api/intelligence/recommendations` in the browser — *"everything the dashboard shows is a JSON API; any other system can consume it."*

10. Questions supervisors commonly ask — and strong answers

"Where does the data come from in a real deployment?" The subsystems (RFID attendance controllers, the leave system, the chatbot) POST canonical events to `/api/events` — authenticated with an API key (timing-safe comparison). The demo uses a seeded simulation of 6 months because real subsystems aren't integrated yet.

"Is the process map hand-drawn?" No — it's a directly-follows graph mined from event sequences (case IDs + timestamps). Layout is automatic (BFS layering). If the real process changes, the map changes on the next mining run. The F&A/RTDD split in the leave process was *discovered*, not configured.

"How are the two leave chains enforced?" One state machine (`src/lib/workflow.ts`): every request needs its supervisor's review first; standard types then route to F&A (admin officer verifies, Director F&A approves), study leave to RTDD (Schedule Officer reviews, Director RTDD approves). Role *and* directorate checks — an RTDD director can't approve annual leave, an F&A officer can't touch study leave.

"How current is the analysis?" A cron trigger runs mining every 6 hours automatically; the dashboard polls every 30 seconds, and the event feed is pushed live over server-sent events. Mining can also be triggered on demand (`POST /api/intelligence/run`).

"What AI models does it use?" Intent classification: Llama 3.2 3B. Policy answers: Llama 3.3 70B, grounded in BGE-base embeddings over Vectorize. Models are config values in `wrangler.jsonc`, swappable. Personal-data queries deliberately bypass the LLM entirely.

"What about security / who can see what?" Machine-to-machine ingestion endpoints require an API key. Per-user identity (Cloudflare Access / SSO) is a planned next phase — the "Acting as" pickers are explicit placeholders for that.

"Can the thresholds be tuned without redeploying?" Yes — bottleneck SLA thresholds live in a KV config namespace (`bottleneck_thresholds_ms`), read at mining time with code defaults as fallback. Changing them is one `wrangler kv key put` command.

"Why Cloudflare Workers?" Serverless at the edge: no server to manage, D1 (SQLite) and Vectorize colocated with the compute, free tier covers the whole demo, and one `wrangler deploy` publishes it. (The repo includes an architecture diagram — `iie_cloudflare_architecture.svg` — if they want the visual.)

"What's the conformance checker actually checking against?" The prescribed leave workflows — the same state machine that enforces live transitions. One source of truth: the model that approves requests is the model that audits them.

"What are the limitations / next phases?" Be honest — it lands well: SSO identity, queue fanout for ingestion bursts, AI narrative over recommendations, and moving the prescribed conformance model into config.

11. Troubleshooting (if something goes wrong mid-demo)

SYMPTOM	FIX
Dashboard blank after clicking a tab	Hard refresh (Ctrl+F5) — you may have a stale cached bundle.
Panels say "Couldn't load ..."	The API is unreachable — check <code>npm run dev</code> is still running; the dashboard retries on its own.
Chatbot says "Something went wrong"	Check internet — Workers AI/Vectorize are live remote services even in local dev. Wait a few seconds and retry (free-tier rate limits).
Stat cards show 0 employees / 0 events	DB is empty: run <code>npm run seed</code> (~3 min), then refresh.
Numbers look doubled (e.g. 66k+ events)	Seed was re-run without <code>--reset</code> . Cosmetic only — or wipe cleanly with <code>npm run seed -- --reset</code> before the demo.
Port 8787 already in use	Another <code>wrangler dev</code> is running — close the old terminal, or just use the already-running one.
<code>npm run validate</code> fails on chatbot metric only	AI free-tier quota (10k neurons/day) — demo the chatbot live instead; it shares the same path.

12. One-minute recap (your closing lines)

"To summarise: one unified event log from every subsystem; automatic discovery of the real process — both leave chains, F&A and RTDD, found by the miner itself; bottleneck and conformance analysis validated at 100% against known ground truth; management-ready recommendations with department comparisons across all nine OHCS units, exportable to CSV and PDF; self-service leave that enforces the real approval rules; and an AI assistant that answers policy from the actual documents and executes real transactions. Everything you saw is running on serverless infrastructure that deploys with a single command."

Generated July 2026 against the seeded dataset (seed 42: 150 staff, 9 units, 33,326 events, 6 months). Numbers quoted are live from the local instance at the time of writing.